Algebraic Identities

Expand & Factorise

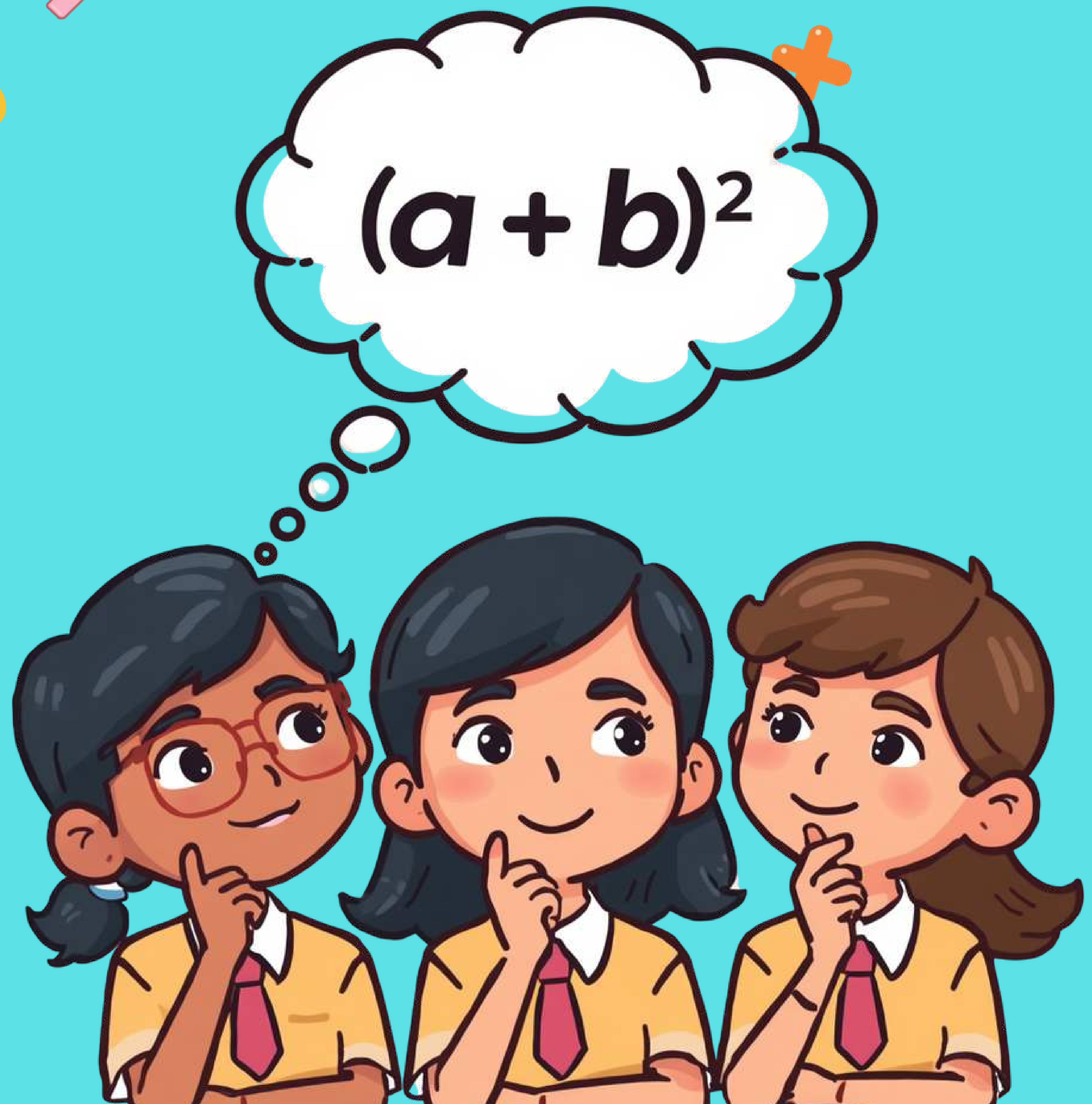
What are

Algebraic Identities?

Algebraic identities are special formulas that help us expand and factorise expressions quickly! They are like magic shortcuts in maths.

Example: $(a + b)^2 = a^2 + 2ab + b^2$

Once you learn these, solving problems becomes super easy!



WHY LEARN ALGEBRAIC IDENTITIES?

They make math easier and faster!

Algebraic identities help us solve problems quickly — like building blocks that fit together perfectly, patterns that repeat, and puzzles that click into place!

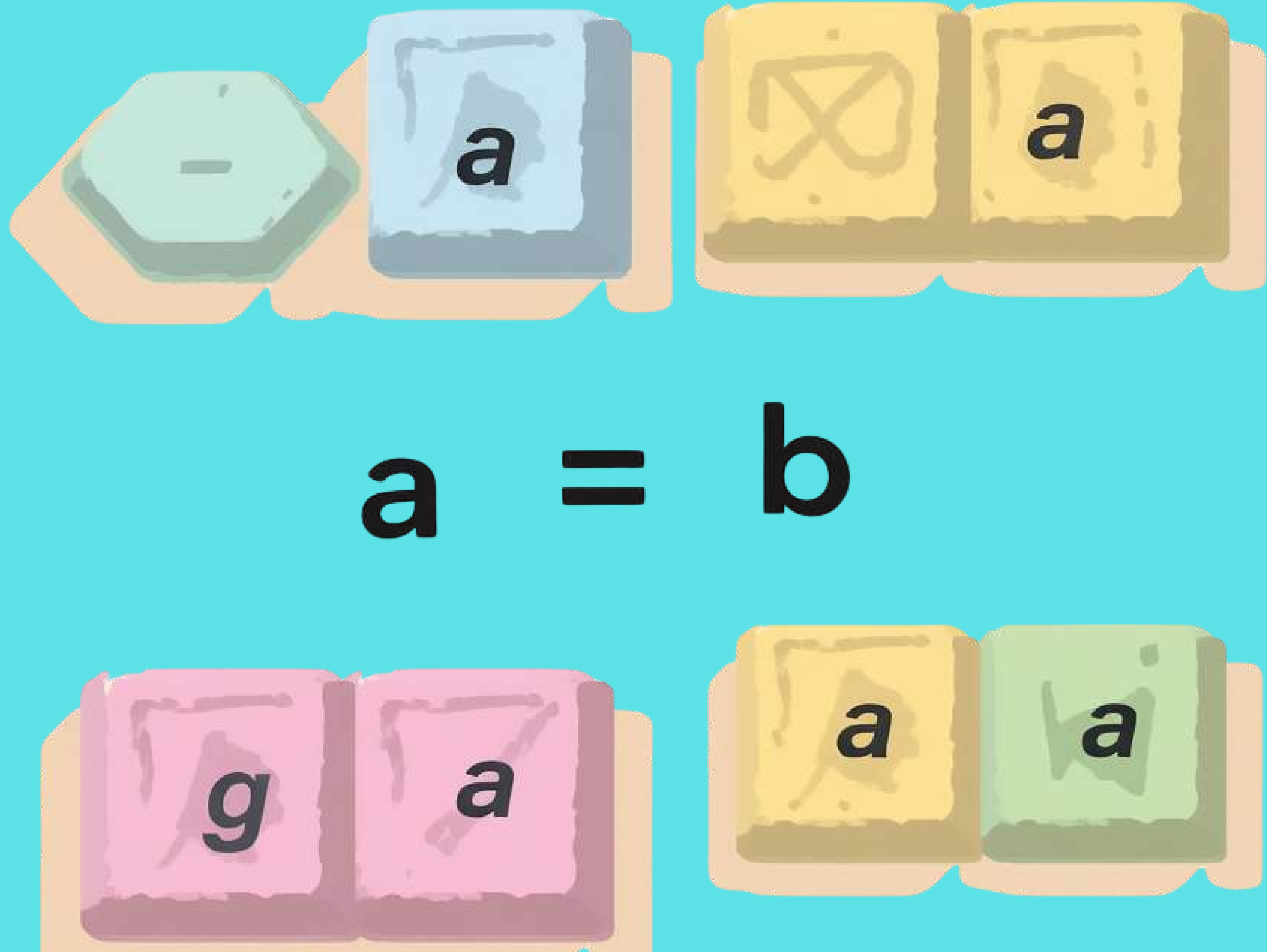
LET'S EXPLORE →

IDENTITY 1:

$(A + B)^2 = A^2 + 2AB + B^2$

Step-by-Step Expansion: $(x + 3)^2$

Step 1: Identify $a = x$ and $b = 3$
Step 2: Apply the formula $(a + b)^2 = a^2 + 2ab + b^2$
Step 3: Substitute: $x^2 + 2(x)(3) + 3^2$
Step 4: Simplify: $x^2 + 6x + 9$
Remember: Square the first, twice the product, square the last!



Identity 2:

$(a - b)^2 = a^2 - 2ab + b^2$

Step-by-Step Example: Expand $(y - 4)^2$

Step 1: Write $(y - 4)^2$ as $(y - 4)(y - 4)$
Step 2: Apply the identity: $a = y$, $b = 4$
Step 3: Calculate each term:
- $a^2 = y^2$
- $2ab = 2 \times y \times 4 = 8y$ (with minus sign: $-8y$)
- $b^2 = 16$

Step 4: Final answer: $y^2 - 8y + 16$

Identity 3:

$a^2 - b^2 = (a + b)(a - b)$

Difference of Squares

This identity helps us factorise when we have the difference of two perfect squares! When you see something like $a^2 - b^2$, you can write it as $(a + b)(a - b)$.

Example: Factorise $9x^2 - 16$

- $9x^2 = (3x)^2$ and $16 = (4)^2$
- So: $9x^2 - 16 = (3x + 4)(3x - 4)$

IDENTITY 4:

(A + B)(A + C)

Expanding Two Binomials

This identity helps us expand two binomials that share a common term!

Formula: $(a + b)(a + c) = a^2 + ac + ab + bc$

Example: Let's expand $(x + 2)(x + 5)$

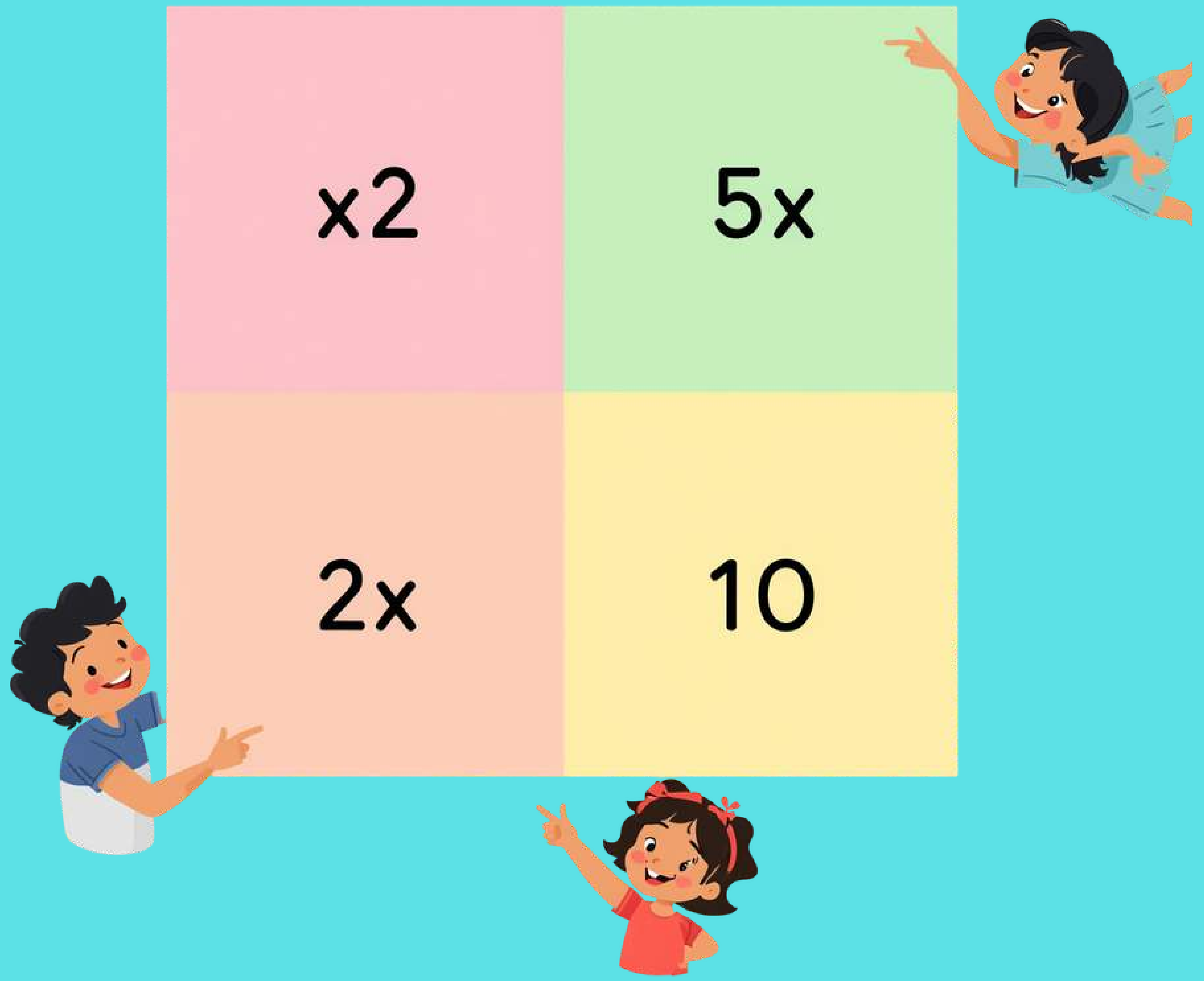
$= x^2 + 5x + 2x + 10$

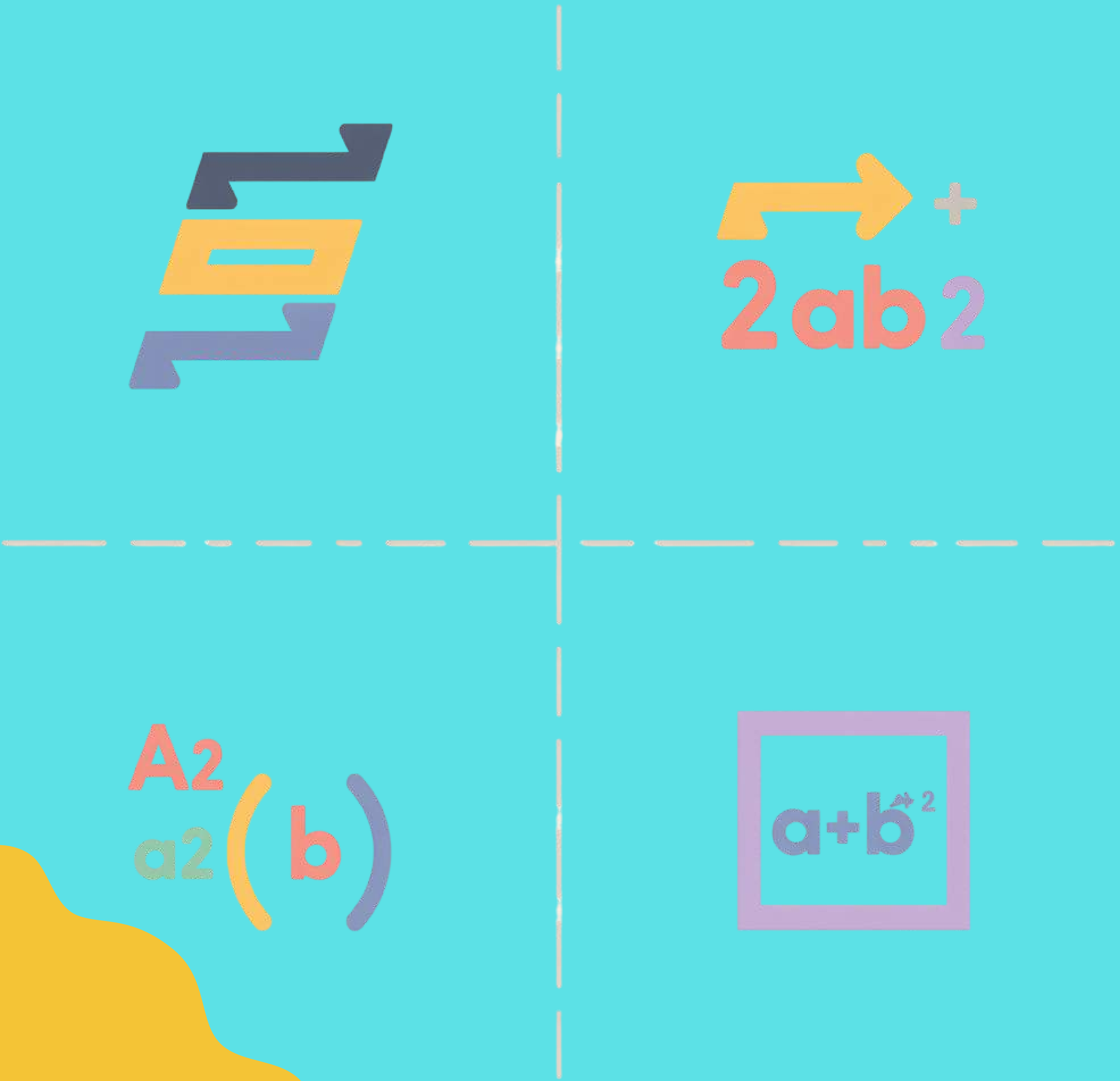
$= x^2 + 7x + 10$

Tip: Use a grid or area model to visualise each term!

Area Model Grids

X = Expansion x = 5





EXPANDING VS FACTORISING

WHAT'S THE DIFFERENCE?

Expanding means opening up the brackets to show all the terms. Factorising means putting expressions back into brackets. Both are useful skills!

PRACTICE

Time!

PRACTICE 1: EXPAND

$$(A + 5)^2 =$$

PRACTICE 2: FACTORISE

$$25x^2 - 9 =$$

PRACTICE 3: EXPAND

$$(x - 3)(x + 4) =$$

Fun Algebra Quiz

What is $(A + B)^2$ equal to?

Key Takeaways

Algebraic identities save time by giving us shortcuts for calculations

Expanding opens brackets, factorising puts expressions back in brackets

You now have powerful tools to solve algebra problems quickly!

Four main identities to remember: $(a+b)^2$, $(a-b)^2$, a^2-b^2, and $(a+b)(a+c)$

Practice makes perfect! The more you practice, the easier it gets

Algebra is fun and useful in everyday life and future studies

Keep Practicing!

You're an Algebra Star!

Great job today! Keep practicing and you'll master all the algebraic identities. Remember: practice makes perfect! Try more problems, challenge yourself, and have fun with math!

THANK YOU!